MolFP-GCN

Learned molecular fingerprints with graph convolutional networks

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The question

Solubility

ESOL dataset

1,128 molecules

Regression on logS

Toxicity

Tox21 dataset

7,823 molecules, 12 assays

17% of labels missing, 7.5% toxic

Explanation

Which atoms drove the prediction?

Saliency colours them in the demo app

Two ways to describe a molecule

Baseline: ECFP + RF

A fixed rule ticks 2048 bits

Same fingerprint whatever the task

Random forest on top

Ours: GCN

Atoms = nodes, bonds = edges

3 rounds of neighbour averaging

Pooled to a 256-number fingerprint, learned for the task

Honest evaluation

Scaffold split

Whole families stay together

Test = unseen structures

80 / 10 / 10 train / val / test

3 seeds

Mean ± std reported

Early stopping on validation

Test set used once

Why it matters

Random split: RMSE 0.80

Scaffold split: RMSE 1.06

The gap is leakage, not skill

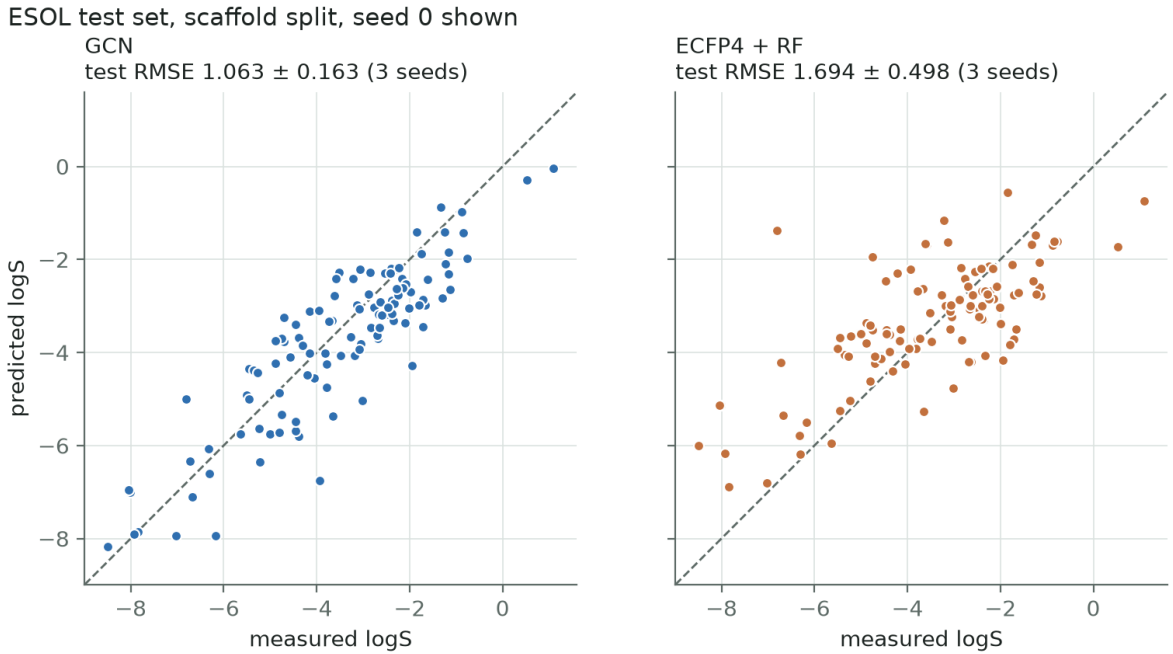
Results

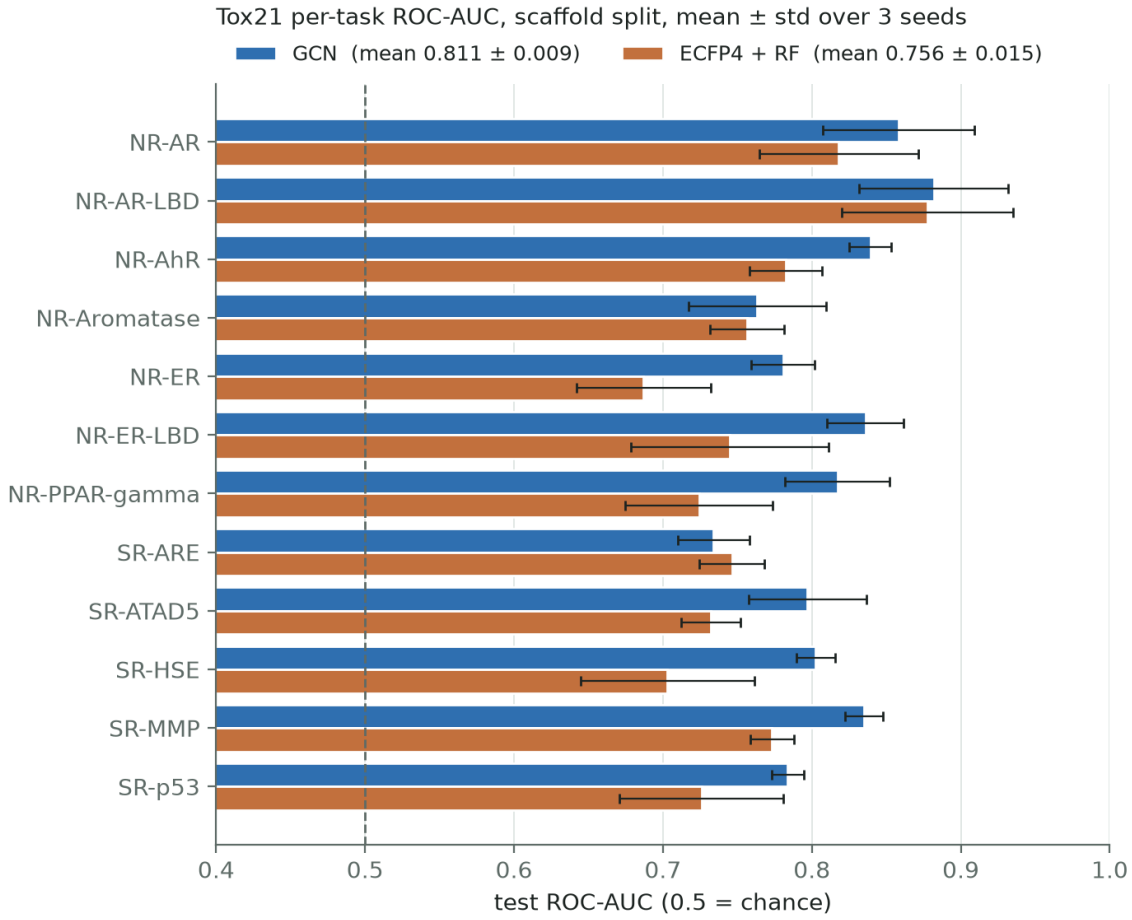
Model	ESOL RMSE ↓	ESOL R^2 ↑	Tox21 ROC-AUC ↑
GCN (learned fingerprint)	**1.063 ± 0.163**	**0.709 ± 0.048**	**0.811 ± 0.009**
ECFP4 + random forest	1.694 ± 0.498	0.267 ± 0.248	0.756 ± 0.015

GCN ahead on 11 of 12 Tox21 assays (RF ahead only on SR-ARE).

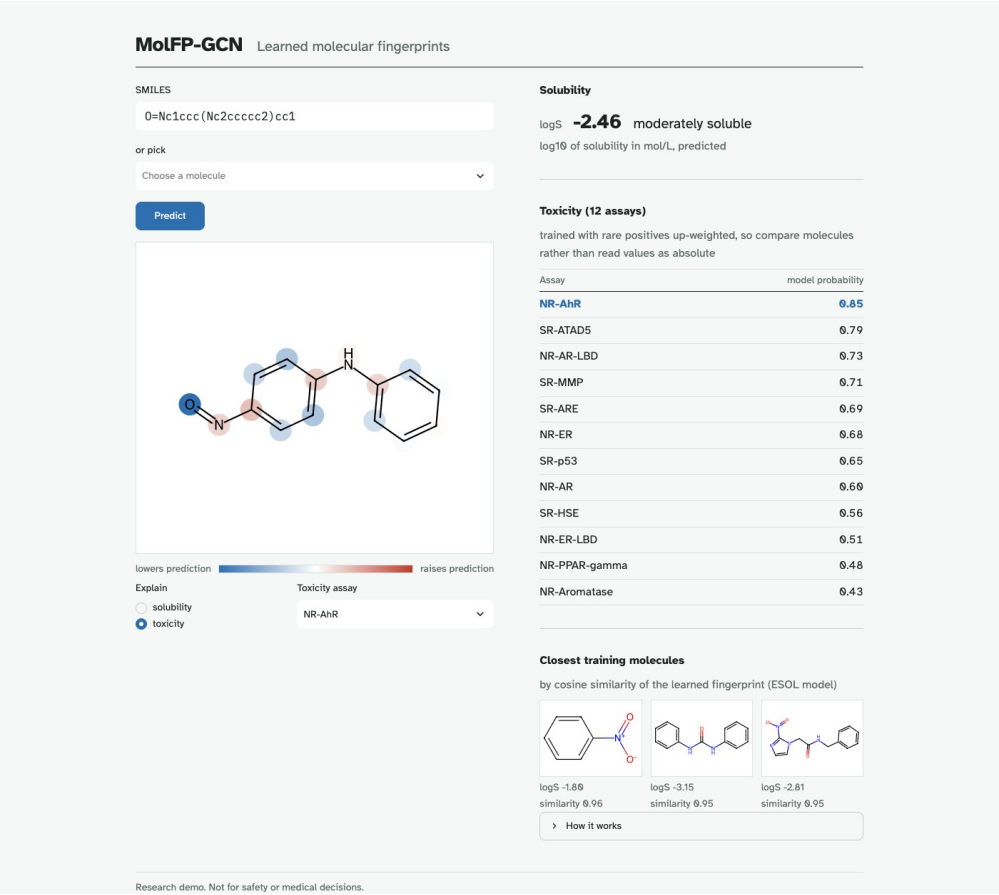
Mean ± std over seeds 0, 1, 2 on a scaffold split.

What the model learned





Explainability and demo



- Draw or paste a SMILES
- Predicts solubility and 12 toxicity scores
- Colours atoms that pushed a prediction up or down
- Run with: make demo
- Or open ?smiles=CCO

Limitations and next steps

Limitations

GCNConv ignores bond types

ESOL is small: big seed-to-seed spread

Scaffold split here may be easier than DeepChem's

Next steps

Edge-aware layers: GINE, D-MPNN

More seeds / cross-validation

Calibrate Tox21 probabilities

Reproducible

make all: 11 min on an M2 CPU

Results identical byte for byte

Tests + 12 invariants

Who did what

Kumaraswamy

GCN architecture

Featurisation: SMILES → graph

Saliency explanations

Simon Peter

Data pipeline and checks

Scaffold split

Training, evaluation, reproducibility

Rakesh

ECFP + RF baseline

Ablations and figures

Demo app, limitations

Thank you

Questions?